

Lecture 6: Proof Notes for Board Work

ISE 5406 – Optimization II, Spring 2026

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These notes contain the proofs and teaching guidance for constrained optimization and KKT conditions.

Green = Steps **Orange** = Key terms **Purple** = Results/Conclusions
Yellow = Write on board *Gray* = Say aloud

Overview: Two central results — (1) KKT conditions are **necessary** at any local minimum (under a constraint qualification), and (2) KKT conditions are **sufficient** for global optimality in convex problems.

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0. Prerequisite: Farkas' Lemma

[Recall from Lecture 3 — this is the key tool for the KKT Necessary proof]

Farkas' Lemma (from Lecture 3)

Let $A \in \mathbb{R}^{m \times n}$ and $\mathbf{c} \in \mathbb{R}^n$. Then **exactly one** of the following has a solution:

System 1: $A\mathbf{x} \leq \mathbf{0}$ and $\mathbf{c}^T \mathbf{x} > 0$ for some $\mathbf{x} \in \mathbb{R}^n$

System 2: $A^T \mathbf{y} = \mathbf{c}$ and $\mathbf{y} \geq \mathbf{0}$ for some $\mathbf{y} \in \mathbb{R}^m$

[Intuition: either you can find a direction $\mathbf{x}$ that all rows of A “approve” ($A\mathbf{x} \leq \mathbf{0}$) but $\mathbf{c}$ doesn't ($\mathbf{c}^T \mathbf{x} > 0$), **or** $\mathbf{c}$ is a non-negative combination of the rows of A . Never both.]

Generalized Form (for mixed inequalities and equalities)

Given vectors $\mathbf{a}_1, \dots, \mathbf{a}_k, \mathbf{b}_1, \dots, \mathbf{b}_l, \mathbf{c} \in \mathbb{R}^n$, **exactly one** holds:

(A) $\exists \mathbf{d}: \mathbf{a}_i^T \mathbf{d} \leq 0$ for all i , $\mathbf{b}_j^T \mathbf{d} = 0$ for all j , $\mathbf{c}^T \mathbf{d} > 0$

(B) $\exists \lambda_i \geq 0, \nu_j \in \mathbb{R}: \mathbf{c} = \sum_i \lambda_i \mathbf{a}_i + \sum_j \nu_j \mathbf{b}_j$

[This is the version we use in the KKT proof with $\mathbf{a}_i = \nabla g_i$, $\mathbf{b}_j = \nabla h_j$, $\mathbf{c} = -\nabla f$]

1. KKT Necessary Conditions

Theorem (KKT Necessary Conditions)

Consider:

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \text{s.t.} \quad g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \quad h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p$$

where f, g_i, h_j are **continuously differentiable**.

If $\bar{\mathbf{x}}$ is a **local minimum** and a **constraint qualification** holds at $\bar{\mathbf{x}}$ (e.g., LICQ), then $\exists \boldsymbol{\lambda}^* \in \mathbb{R}^m, \boldsymbol{\nu}^* \in \mathbb{R}^p$ such that:

- (a) **Stationarity:** $\nabla f(\bar{\mathbf{x}}) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\bar{\mathbf{x}}) + \sum_{j=1}^p \nu_j^* \nabla h_j(\bar{\mathbf{x}}) = 0$
- (b) **Primal Feasibility:** $g_i(\bar{\mathbf{x}}) \leq 0$ for all i ; $h_j(\bar{\mathbf{x}}) = 0$ for all j
- (c) **Dual Feasibility:** $\lambda_i^* \geq 0$ for all i
- (d) **Complementary Slackness:** $\lambda_i^* g_i(\bar{\mathbf{x}}) = 0$ for all i

Proof

[The proof has four steps: define active set, show no feasible descent, apply Farkas, use CQ to normalize]

Step 1: Define the Active Constraint Set

$$\mathcal{A}(\bar{\mathbf{x}}) := \{i : g_i(\bar{\mathbf{x}}) = 0\}$$

[These are the constraints that are “tight” at the optimum]

For $i \notin \mathcal{A}(\bar{\mathbf{x}})$: $g_i(\bar{\mathbf{x}}) < 0$ (**slack/inactive**)

Step 2: Show No Feasible Descent Direction Exists

[This is the geometric core of the proof]

Claim: $\nexists \mathbf{d} \in \mathbb{R}^n$ such that **all three** hold:

- 1. $\nabla f(\bar{\mathbf{x}})^T \mathbf{d} < 0$ *[descent in objective]*
- 2. $\nabla g_i(\bar{\mathbf{x}})^T \mathbf{d} \leq 0$ for all $i \in \mathcal{A}(\bar{\mathbf{x}})$ *[stays feasible for active constraints]*
- 3. $\nabla h_j(\bar{\mathbf{x}})^T \mathbf{d} = 0$ for all $j = 1, \dots, p$ *[stays on equality constraints]*

Suppose such a $\mathbf{d}$ existed. By C^1 smoothness, for small $t > 0$:

$$\begin{aligned}
f(\bar{\mathbf{x}} + t\mathbf{d}) &\approx f(\bar{\mathbf{x}}) + t\nabla f(\bar{\mathbf{x}})^T \mathbf{d} < f(\bar{\mathbf{x}}) \\
g_i(\bar{\mathbf{x}} + t\mathbf{d}) &\approx \underbrace{g_i(\bar{\mathbf{x}})}_{=0} + t \underbrace{\nabla g_i(\bar{\mathbf{x}})^T \mathbf{d}}_{\leq 0} \leq 0 \quad (i \in \mathcal{A}) \\
g_i(\bar{\mathbf{x}} + t\mathbf{d}) &< 0 \quad (i \notin \mathcal{A}) \quad [\text{by continuity, for small } t] \\
h_j(\bar{\mathbf{x}} + t\mathbf{d}) &\approx \underbrace{h_j(\bar{\mathbf{x}})}_{=0} + t \underbrace{\nabla h_j(\bar{\mathbf{x}})^T \mathbf{d}}_{=0} = 0
\end{aligned}$$

$\Rightarrow$ Feasible point with lower objective $\Rightarrow$ contradicts $\bar{\mathbf{x}}$ being a local minimum

No feasible descent direction exists at $\bar{\mathbf{x}}$ ✓

Step 3: Apply Farkas' Lemma / Fritz John Conditions

[Farkas' Lemma converts "no feasible descent direction" into algebraic multipliers]

[Step 2 says System (A) from the generalized Farkas has no solution (with $\mathbf{a}_i = \nabla g_i$, $\mathbf{b}_j = \nabla h_j$, $\mathbf{c} = -\nabla f$). So System (B) must hold.]

Since no such $\mathbf{d}$ exists, by **Farkas' Lemma**, $\exists$:

- $\lambda_0 \geq 0$
- $\lambda_i \geq 0$ for $i \in \mathcal{A}(\bar{\mathbf{x}})$, $\lambda_i = 0$ for $i \notin \mathcal{A}(\bar{\mathbf{x}})$
- $\nu_j \in \mathbb{R}$ (unrestricted) for $j = 1, \dots, p$

$$\lambda_0 \nabla f(\bar{\mathbf{x}}) + \sum_{i \in \mathcal{A}} \lambda_i \nabla g_i(\bar{\mathbf{x}}) + \sum_{j=1}^p \nu_j \nabla h_j(\bar{\mathbf{x}}) = 0$$

with not all $\lambda_0, \lambda_i, \nu_j$ zero (**Fritz John conditions**)

Step 4: Apply Constraint Qualification to Normalize

[CQ is essential — without it, we might get the trivial case $\lambda_0 = 0$]

LICQ requires: $\{\nabla g_i(\bar{\mathbf{x}}) : i \in \mathcal{A}\} \cup \{\nabla h_j(\bar{\mathbf{x}}) : j = 1, \dots, p\}$ are **linearly independent**

$$\text{If } \lambda_0 = 0: \sum_{i \in \mathcal{A}} \lambda_i \nabla g_i + \sum_j \nu_j \nabla h_j = 0 \text{ with not all zero}$$

$\Rightarrow$ Contradicts LICQ (linear independence)

$\Rightarrow \lambda_0 \neq 0$, so we can set $\lambda_0 = 1$

Define $\lambda_i^* := \lambda_i$, $\nu_j^* := \nu_j$:

$$\nabla f(\bar{\mathbf{x}}) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\bar{\mathbf{x}}) + \sum_{j=1}^p \nu_j^* \nabla h_j(\bar{\mathbf{x}}) = 0$$

[This is stationarity. Conditions (b)–(d) follow from the construction.]

KKT Necessary Conditions proved □

Teaching Notes

- **Core Insight:** The proof is fundamentally **geometric**. At a local min, the descent cone and feasible cone are disjoint. Farkas' Lemma converts this to algebra.
- **Board Diagram:** Draw 2D constraint set with $\bar{\mathbf{x}}$ on boundary. Show ∇f pointing away from feasible region. Show the feasible cone vs. descent cone are disjoint.
- **Farkas' Lemma is the workhorse:** Students should understand it as: “if a system of linear inequalities has no solution, then a dual certificate exists.”
- **Role of CQ:** Without LICQ, we only get **Fritz John conditions** (λ_0 might be 0). CQ ensures $\lambda_0 \neq 0$ so we can normalize.
- **Time estimate:** 10–15 minutes on board (outline key steps, work Step 2 in detail).

Common Student Confusions

1. **“Why $\lambda_i^* \geq 0$?”** For min with $g_i \leq 0$: positive λ_i means tightening the constraint worsens the objective. If $\lambda_i < 0$, weakening the constraint would *improve* the solution — contradicting optimality.
2. **“What if I don't know which constraints are active?”** That's what complementary slackness tells you! Either $\lambda_i^* = 0$ (inactive) or $g_i(\bar{\mathbf{x}}) = 0$ (active). When solving, guess the active set and verify.
3. **“Why continuous differentiability?”** Step 2 uses Taylor (linear) approximations. Without C^1 , these don't hold. Non-smooth problems need subdifferential calculus.
4. **“What is LICQ exactly?”** At $\bar{\mathbf{x}}$, the gradients $\{\nabla g_i(\bar{\mathbf{x}}) : i \in \mathcal{A}\} \cup \{\nabla h_j(\bar{\mathbf{x}}) : j = 1, \dots, p\}$ must be **linearly independent**. Intuition: active constraint boundaries don't “run parallel” at the optimum.

2. KKT Sufficient Conditions for Convex Problems

Theorem (KKT Sufficiency for Convex Problems)

Same problem as above. Assume f and $g_1, \dots, g_m$ are **convex**, and $h_1, \dots, h_p$ are **affine**.
If $\exists \bar{\mathbf{x}}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*$ satisfying KKT conditions (a)–(d), then $\bar{\mathbf{x}}$ is a **global minimum**.

Proof

[This is the proof to fully develop on the board. It's the more important one for students to see.]

Goal: Show $f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$ for any feasible $\mathbf{x}$

Step 1: Convexity and First-Order Inequality

[This is the key property of convex functions]

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}})$$

[The tangent plane underestimates a convex function — this is the definition of convexity]

Step 2: Substitute Stationarity Condition

From condition (a):

$$\nabla f(\bar{\mathbf{x}}) = -\sum_{i=1}^m \lambda_i^* \nabla g_i(\bar{\mathbf{x}}) - \sum_{j=1}^p \nu_j^* \nabla h_j(\bar{\mathbf{x}})$$

Substituting into Step 1:

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) - \sum_i \lambda_i^* \nabla g_i(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) - \sum_j \nu_j^* \nabla h_j(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}})$$

[Now we need to show both sums are ≥ 0]

Step 3: Use Convexity of Inequality Constraints

Since each g_i is **convex**:

$$g_i(\mathbf{x}) \geq g_i(\bar{\mathbf{x}}) + \nabla g_i(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}})$$

Rearranging: $\nabla g_i(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq g_i(\mathbf{x}) - g_i(\bar{\mathbf{x}})$

Step 4: Use Dual Feasibility and Feasibility

For any feasible $\mathbf{x}$:

- **Primal feasibility**: $g_i(\bar{\mathbf{x}}) \leq 0$ and $g_i(\mathbf{x}) \leq 0$
- **Dual feasibility**: $\lambda_i^* \geq 0$

[This is where dual feasibility becomes crucial]

Since $\lambda_i^* \geq 0$ and $g_i(\mathbf{x}) \leq 0$, combined with **complementary slackness** ($\lambda_i^* g_i(\bar{\mathbf{x}}) = 0$):

$$-\lambda_i^* \nabla g_i(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \geq -\lambda_i^* g_i(\mathbf{x}) + \underbrace{\lambda_i^* g_i(\bar{\mathbf{x}})}_{=0} = -\lambda_i^* g_i(\mathbf{x}) \geq 0$$

[Non-negative times non-positive gives non-positive, so the negative is non-negative]

Summing over all i :

$$-\sum_{i=1}^m \lambda_i^* \nabla g_i(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \geq 0$$

Inequality constraint terms contribute ≥ 0 ✓

Step 5: Use Affinity of Equality Constraints

Since h_j are **affine**: $\nabla h_j(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) = h_j(\mathbf{x}) - h_j(\bar{\mathbf{x}})$

[This identity holds exactly because h_j is affine — it fails for nonlinear equalities!]

Feasibility: $h_j(\mathbf{x}) = 0$ and $h_j(\bar{\mathbf{x}}) = 0$

$$-\sum_j \nu_j^* \nabla h_j(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) = -\sum_j \nu_j^* \cdot 0 = 0$$

Equality constraint terms contribute exactly 0 ✓

Step 6: Combine All Inequalities

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) - \underbrace{\sum_i \lambda_i^* \nabla g_i^T(\bar{\mathbf{x}}) (\mathbf{x} - \bar{\mathbf{x}})}_{\geq 0 \text{ (Step 4)}} - \underbrace{\sum_j \nu_j^* \nabla h_j^T(\bar{\mathbf{x}}) (\mathbf{x} - \bar{\mathbf{x}})}_{= 0 \text{ (Step 5)}} \geq f(\bar{\mathbf{x}})$$

$\Rightarrow f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$ for any feasible $\mathbf{x}$

$\bar{\mathbf{x}}$ is a global minimum □

Teaching Notes

- **Key Insight:** The entire proof is a chain of inequalities powered by **convexity**. Without convexity, Step 1 fails.
- **Board Walkthrough:** Draw a 2D convex feasible region with level sets. Mark $\bar{\mathbf{x}}$. Then walk through the algebra step by step. Emphasize each “ $\geq$ ” and where it comes from.
- **Dual Feasibility’s Role:** If $\lambda_i^* < 0$, the inequality in Step 4 *reverses*. This is why $\lambda_i^* \geq 0$ is not arbitrary — it’s structurally necessary.
- **Why Affine Equalities?** Non-affine h_j breaks Step 5. Give the counterexample: $h(x) = x_1^2 + x_2^2 - 1$ (unit circle). At $\bar{\mathbf{x}} = (1, 0)$, $\mathbf{x} = (0, 1)$: $\nabla h^T(\mathbf{x} - \bar{\mathbf{x}}) = -2 \neq h(\mathbf{x}) - h(\bar{\mathbf{x}}) = 0$.
- **Global vs. Local:** Theorem 1 gives *local* necessity. Theorem 2 gives *global* sufficiency. This asymmetry is the power of convexity!
- **Time estimate:** 12–15 minutes for full board walkthrough.

Common Student Confusions

1. **“Why must equality constraints be affine?”** Non-affine equalities don’t satisfy $\nabla h^T(\mathbf{x} - \bar{\mathbf{x}}) = h(\mathbf{x}) - h(\bar{\mathbf{x}})$. The proof breaks at Step 5.
2. **“What if some $\lambda_i^* < 0$?”** Then KKT condition (c) is violated, so the theorem doesn’t apply. The inequality in Step 4 reverses.
3. **“Why does the proof work forward (KKT $\Rightarrow$ optimal) but not backward?”** Forward uses convexity. Backward is Theorem 1, which needs CQ but *not* convexity. KKT is necessary in general, sufficient for convex.
4. **“If g_i is not convex, does KKT fail to be sufficient?”** Yes. Non-convex constraints can create local minima where KKT holds but the point is not globally optimal.
5. **“Do we need the Lagrangian?”** It’s implicit. Stationarity says $\nabla_{\mathbf{x}}\mathcal{L} = 0$, and the proof is equivalent to showing the Lagrangian is minimized at $\bar{\mathbf{x}}$.

3. Alternative Proof: Lagrangian Approach (Optional)

Alternative (can mention as supplement — same hypotheses as Theorem 2)

Define the Lagrangian:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i^* g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j^* h_j(\mathbf{x})$$

Key observations:

1. For feasible $\mathbf{x}$ with $\boldsymbol{\lambda} \geq 0$: $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f(\mathbf{x})$ *[since $\lambda_i g_i(\mathbf{x}) \leq 0$, $\nu_j h_j(\mathbf{x}) = 0$]*
 2. If KKT holds at $(\bar{\mathbf{x}}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$: $\mathcal{L}(\bar{\mathbf{x}}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = f(\bar{\mathbf{x}})$ *[by complementary slackness]*
 3. Since $\nabla_{\mathbf{x}} \mathcal{L} = 0$ and $\mathcal{L}$ is convex in $\mathbf{x}$: $\mathcal{L}(\bar{\mathbf{x}}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \leq \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$
- $\Rightarrow f(\bar{\mathbf{x}}) = \mathcal{L}(\bar{\mathbf{x}}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \leq \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \leq f(\mathbf{x})$ for all feasible $\mathbf{x}$

4. Connections to Other Results

- **Unconstrained Optimization:** Set $m = p = 0$. KKT reduces to $\nabla f(\bar{\mathbf{x}}) = 0$.
- **Lagrange Multipliers (Equality Only):** Set $m = 0$. KKT becomes the classical method with ν_j (any sign).
- **Slater's Condition:** For convex problems, $\exists \tilde{\mathbf{x}}$ with $g_i(\tilde{\mathbf{x}}) < 0$ strictly for all i implies a CQ holds. Often easier to verify than LICQ.
- **Duality Theory (Lecture 7):** The multipliers $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ define the dual problem. For convex problems, **strong duality** holds: $p^* = d^*$. KKT sufficiency can be seen as a consequence of strong duality.
- **Penalty / Augmented Lagrangian Methods:** The Lagrangian $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ connects directly to penalty and augmented Lagrangian algorithms used in practice.
- **Interior Point Methods:** Modern solvers find KKT points iteratively. Theorem 2 justifies why finding a KKT point = finding the global optimum (for convex problems).

5. Summary Table: KKT Properties

Property	General Problem	Convex Problem
KKT is Necessary	Yes (with CQ)	Yes (with CQ)
KKT is Sufficient	No	Yes
Optimum is Local	Yes (necessary)	N/A (sufficient)
Optimum is Global	N/A (necessary)	Yes (sufficient)

CQ = Constraint Qualification (e.g., LICQ, Slater's condition)

6. Proof Roadmap

Theorem	Key Technique	Est. Time	Priority
KKT Necessary	No feasible descent + Farkas' Lemma	10–15 min	Outline
KKT Sufficient	Convexity + inequality chain	12–15 min	Full proof

Dependency: The necessary conditions proof uses **Farkas' Lemma** (from Lecture 3). The sufficient conditions proof is self-contained, relying only on the definition of convexity.

Total estimated board time: 22–30 minutes.

Recommendation: Do KKT Sufficient (Theorem 2) in **full** on the board — it is the more important proof for students to see worked out. For KKT Necessary (Theorem 1), give a proof **outline** on slides and emphasize Steps 2 and 3.

7. Practice Problems for Students

KKT Necessary Conditions

1. **Identifying Active Constraints:** Consider $\min f(\mathbf{x})$ s.t. $g_1(\mathbf{x}) = x_1^2 + x_2^2 - 4 \leq 0$, $g_2(\mathbf{x}) = -x_1 - 1 \leq 0$, at $\bar{\mathbf{x}} = (2, 0)$. Which constraints are active? Write the KKT conditions (without solving).
2. **Lagrange Multipliers vs. KKT:** Write the KKT conditions for $\min f(\mathbf{x})$ s.t. $h(\mathbf{x}) = x_1^2 + x_2^2 - 1 = 0$ (equality only). Compare to the method of Lagrange multipliers.
3. **Constraint Qualification:** Consider $\min x_1$ s.t. $g_1(\mathbf{x}) = x_1^2 - x_2 \leq 0$, $g_2(\mathbf{x}) = x_1 \leq 0$, at $\bar{\mathbf{x}} = (0, 0)$. Compute gradients of active constraints. Does LICQ hold?

KKT Sufficient Conditions

1. **Convex Quadratic:** Show that $\min \frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} - \mathbf{b}^T \mathbf{x}$ s.t. $\mathbf{x} \geq 0$ has a global optimum at any $\bar{\mathbf{x}}$ satisfying KKT (assume $\mathbf{Q} \succ 0$).
2. **Linear Program:** Write the KKT conditions for an LP and verify sufficiency via Theorem 2.
3. **Non-convex Failure:** Construct an example where a non-convex function satisfies KKT at a point but is not globally optimal.

Appendix: Notation

Symbol	Meaning
$\mathbb{R}^n$	Euclidean space of dimension n
$\mathbf{x}$	Point in the feasible region
$\bar{\mathbf{x}}$	Local/global minimum (as applicable)
$f(\mathbf{x})$	Objective function
$g_i(\mathbf{x}) \leq 0$	Inequality constraints
$h_j(\mathbf{x}) = 0$	Equality constraints
λ_i^*	Lagrange multiplier for inequality constraint i
ν_j^*	Lagrange multiplier for equality constraint j
$\nabla f(\mathbf{x})$	Gradient of f at $\mathbf{x}$
$\mathcal{A}(\mathbf{x})$	Set of active inequality constraints at $\mathbf{x}$
$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$	Lagrangian function
LICQ	Linear Independence Constraint Qualification
CQ	Constraint Qualification (generic)